
Revision Table

Revision No.	Details	Section / Page No.
1	Initial thesis structure setup	All chapters
2	Added LaTeX packages for code and formatting	Preamble
3	Restructured to match approved ClassiFish format	Chapter organization
4	[Future revisions to be added here]	[Location]

Table 1: Thesis Revision History



Consol: A Notes-to-Recollection Application Using SimCSE for its Textual Semantic Comparison

George Christopher C. Daguman

Supervised by
Dante D. Dinawanao

Department of Computer Science
College of Computer Studies
MSU - Iligan Institute of Technology

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Abstract

Generative AI has presented society with an opportunity to create a surplus of educational content in mere seconds. While knowledge is inherently beneficial and promotes intellectual and academic growth, there still is a bridge to actual learning that cannot be crossed just through using AI and its instantaneous generative abilities (Mishra & Anthony, 2023). Learning, in the academic landscape, is comprised of obtaining information which can be in the form of notes (Rashid & Rigas, 2006; Devlin et al., 2019). Retaining knowledge after note-taking remains a challenge, as forgetting is inevitable and overlooked by recent AI and NLP advancements. (Klemm, 2012; Naim et al., 2020). In studying and reviewing, if a person wishes to ensure their recollection of key topics is accurate and coherent, options are limited without manually referencing source material, which can be tiring and inefficient. Assistance is needed, particularly in providing instant feedback for accuracy and mastery of notes (Karpicke & Roediger, 2008; Sachs, 1967). This research proposes Consol, a notes-to-recollection application that uses SimCSE's contrastive sentence embeddings and semantic similarity analysis to compare how similar a person's recollection is to their original notes (Gao et al., 2021; Corley et al., 2005), and uses progress tracking techniques to encourage consistency and improvement (Deterding et al., 2011; Gaston & Cooper, 2017). It will be a web-based application that leverages semantic similarity evaluation to assess memory recall effectively. The study aims to evaluate whether semantic-based feedback enhances learning outcomes compared to traditional recall methods. Through system testing and user performance assessment, this research will analyze the system's usability, accuracy, and effectiveness in improving memory retention (Baars et al., 2022; Tran & Nguyen, 2022). The results aim to provide a scalable solution bridging note-taking and active learning while fostering consistent knowledge consolidation (Vişcu, 2024; Stojanovic et al., 2020).

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Introduction

Memory is not merely the result of recording information but the outcome of consistent review and active engagement with what has been learned. Notes, while essential for capturing ideas, do not inherently lead to long-term retention or understanding simply by being created. Instead, it is through revisiting, processing, and consolidating these notes that individuals truly reinforce their knowledge (Karpicke and Roediger, 2008; Klemm, 2012).

For students, teachers, and professionals alike, the challenge lies in establishing consistent learning practices and receiving validation for their efforts—validation that not only measures recall but also provides meaningful feedback on comprehension (Stojanovic et al., 2020; Baars et al., 2022). Traditional note-taking methods often focus on transcription, prioritizing the exact wording of information over its deeper meaning. However, research by Sachs (1967) highlights that individuals naturally retain the semantic meaning—the "gist" of information—rather than its verbatim form.

This tendency to paraphrase and restructure ideas reflects the cognitive process of abstraction, where understanding takes precedence over memorization. Conventional systems that evaluate recall by word-for-word accuracy fail to account for this natural process, offering an incomplete and often discouraging measure of learning performance (Corley et al., 2005; Gao et al., 2021).

Additionally, progress tracking has proven to be an effective strategy for sustaining motivation and fostering consistent learning habits. By incorporating elements such as performance metrics, mastery indicators, and progress tracking, such systems provide users with actionable insights and a sense of achievement (Deterding et al., 2011; Gaston and Cooper, 2017). These tools not only validate the user's learning progress but also

make the process more interactive and rewarding, promoting deeper engagement and long-term retention (Tran and Nguyen, 2022; Viscu, 2024).

1.1 | Background of the Study

Now, as technology continues to advance in the current “information age,” the demand for fast, accurate, and easily accessible information has increased dramatically. This is particularly evident in fields where knowledge acquisition is critical, such as education, research, and personal development. The proliferation of generative AI technologies and digital tools has revolutionized the ways individuals gather, process, and manage information. Digital note-taking tools, such as Evernote, Notion, Obsidian, and various AI-assisted applications, offer features like automatic transcription, cloud synchronization, and summarization, making the recording of ideas and knowledge more efficient and convenient to the modern user (Baars et al., 2022; Rashid and Rigas, 2006).

Existing studies confirm that effective learning requires more than passive note creation. Research conducted by Sachs (1967) indicates that individuals are better at retaining semantic meaning rather than exact syntactic details. This means that during memory retrieval, individuals often paraphrase, restructure, and abstract ideas rather than reproduce content verbatim. Traditional systems that rely on word-for-word accuracy to evaluate recall fail to account for this natural cognitive process, leading to an incomplete understanding of memory performance (Corley et al., 2005; Gao et al., 2021).

Building on this, the integration of AI-driven systems in learning environments has opened new avenues for assessing memory and comprehension more effectively. By including the more recent advancements in natural language processing (NLP) and similarity-based algorithms, these systems can evaluate recall beyond mere verbatim reproduction, focusing instead on the semantic alignment between original and recalled content (Devlin et al., 2019; Gao et al., 2021). Such approaches recognize the inherent flexibility of human cognition, where meaningful understanding takes precedence over exact replication. For example, sentence-transformer models like SimCSE have demonstrated the ability to measure semantic similarity with high accuracy, making them suitable for applications that assess memory recall in a way that mirrors human evaluation (Gao et al., 2021; Corley et al., 2005).

This shift toward semantic-based assessment not only provides a more accurate

measure of knowledge retention but also aligns with modern pedagogical principles that emphasize deep learning and critical thinking. By rewarding comprehension and conceptual mastery over rote memorization, AI-assisted systems encourage learners to engage more actively with the material, fostering long-term knowledge retention and improved cognitive performance (Karpicke and Roediger, 2008; Klemm, 2012). As these technologies continue to evolve, they hold the potential to transform traditional methods of learning evaluation and pave the way for more adaptive and personalized educational tools (Stojanovic et al., 2020; Viscu, 2024).

1.2 | Statement of the Problem

The rapid development and advancement of Generative AI has led to the overwhelming abundance of knowledge, and with that, an unlimited opportunity to take notes. While this is clearly for some benefit, it is not put into actual use if the increasing number of notes are not consolidated into memory, thereby hindering the cumulative learning process (Karpicke & Roediger, 2008; Klemm, 2012). Additionally, in individual study contexts, there is a lack of mechanisms to verify whether a learner has truly understood or mastered the material. This problem is compounded by the fact that human recall often involves paraphrasing and abstracting information, making it difficult for current systems to accurately assess a learner's understanding (Sachs, 1967; Corley et al., 2005). Consequently, there is a need for a system that can evaluate learning by recognizing and comparing both the original content and its reconstructed ideas through context clues, ensuring that abstracted yet relevant concepts are acknowledged without overlooking their presence in the material (Gao et al., 2021; Devlin et al., 2019).

1.3 | Objectives

1.3.1 | General Objective

To develop a web-based memorization system that refers to the user's pre-made notes and prompts the user to perform active textual recall, leveraging SimCSE to maintain fairness during comparison despite non-verbatim phrasing of ideas.

1.3.2 | Specific Objectives

1. Design the system architecture of the notes-to-recollection comparison system, including its scoring criteria for evaluating recall accuracy.
2. Devise pre-processing procedures for removing unnecessary non-alphanumeric symbols and formatting artifacts to be suitable for semantic comparison.
3. Employ Simple Contrastive Sentence Embeddings (SimCSE) for semantic comparison of two given texts.
4. Implement a web-based application integrating the SimCSE Model, giving it an appropriate user experience that facilitates motivation through progress tracking and active learning.
5. Conduct a usability test on the web-based application and document the results.

1.4 | Scope and Limitation

The study is strictly limited to a maximum of 10 test participants. The study will be conducted from January 2025 through April of the same year.

1.5 | Significance and Contribution

The proposed implementation will assist users in memorizing their notes and incentivize steady progress in content mastery through various tracking strategies.

Review of Related Literature

This chapter reviews relevant literature on note-taking strategies, memory processes, natural language processing (NLP) for semantic similarity, and progress tracking techniques. These areas provide the theoretical basis for understanding information encoding, memory consolidation, textual comparison, and user engagement. By examining these interconnected concepts, the review establishes a foundation for exploring methods that enhance learning, recall, and performance. The studies discussed offer insights into optimizing knowledge retention, evaluating recall accuracy, and motivating users through structured feedback mechanisms.

2.1 | Memory and Cognitive Processes

Memory plays a crucial role in the learning process, specifically in the retention and recall of information. Studies have demonstrated that memory prioritizes the retention of meaning over surface-level details, which has direct implications for comprehension and learning.

2.1.1 | The Role of Semantic and Syntactic Memory

Sachs (1967) explored the relationship between semantic and syntactic memory in the context of comprehension. In her study, participants were presented with sentences and tested on their ability to recognize changes in meaning and form after brief intervals. The findings revealed that memory retention is primarily focused on semantic content, while syntactic details, such as active-to-passive transformations, fade quickly. Even when participants retained the overall meaning of the sentences, the formal properties of the sentences were quickly forgotten. This study concludes that memory is structured

to prioritize the “deep structure” or meaning of a sentence over its “surface structure” or syntax, suggesting that comprehension relies on extracting and retaining the core meaning of information.

2.1.2 | Cognitive Processes in Memory Retention

Rashid and Rigas (2006) provide an analysis of memory stages and their relationship to learning and information retention. The study outlines the progression of information through sensory memory, short-term memory, and long-term memory, highlighting the necessity of active cognitive processing for effective retention. Encoding, which involves the initial capture of relevant information, and reviewing, which reinforces the transfer of information to long-term memory, are identified as critical phases in the learning process. This study also discusses retrieval failure as a cause of forgetting, where proactive interference and retroactive interference disrupt recall.

Proactive interference occurs when previously learned information impairs the learning of new material, while retroactive interference arises when newly acquired knowledge interferes with the recall of older information. These findings emphasize that memory retention is optimized when learners engage in deep processing activities, such as organizing and summarizing information, rather than relying on surface-level strategies. This perspective aligns with the findings of Sachs (1967), reinforcing the importance of meaningful encoding and the prioritization of core content over superficial details.

2.1.3 | Memory and Recall

Naim et al. (2020) identified that recall performance grows sublinearly as the list length increases. This sublinear growth highlights a saturation effect, where the number of items retrieved plateaus despite an increase in encoded material. The study also revealed that encoding quality plays a crucial role in determining recall performance. Slower presentation rates allowed participants to encode a higher number of words into memory, which improved the total recall capacity. These findings provide evidence that memory recall adheres to universal laws, offering a structured and deterministic perspective on retrieval processes.

2.1.3.1 | Retrieval-Based Testing

Karpicke and Roediger (2008) emphasize that testing is not merely a means of assessment but a potent strategy for enhancing long-term memory retention. Their study debunks the conventional belief that once information is initially learned, further practice adds little value. Instead, repeated retrieval—through testing—promotes deeper encoding and strengthens memory consolidation, highlighting retrieval as an active mechanism essential for sustained learning performance.

2.1.3.2 | Optimizing Retrieval Practice for Long-term Retention

Karpicke and Roediger (2008) demonstrated that retrieval practice plays a central role in consolidating memories and improving long-term retention. Their experiments revealed that repeated retrieval significantly outperforms repeated study in enhancing memory performance. This finding validates the importance of active recall over passive encoding strategies.

2.2 | Note-Taking Techniques and Tools

Note-taking serves as an essential tool for facilitating memory retention and comprehension. Various techniques and tools have been developed to assist learners in organizing and processing information effectively.

2.2.1 | Traditional Note-Taking Techniques

Rashid and Rigas (2006) examine the comparative effectiveness of traditional note-taking methods, including the Cornell Method, the Outlining Method, and the Mapping Method. These methods provide learners with strategies for organizing content to improve comprehension and retention.

2.2.2 | Digital Note-Taking and E-Learning Tools

Rashid and Rigas (2006) further explore the role of technology in enhancing note-taking and memory retention within e-learning environments. The integration of tools such as Microsoft OneNote and Tablet PCs has provided learners with new ways to engage with content, enabling multimedia annotations and portable accessibility.

2.2.2.1 | Quizlet

Tran and Nguyen (2022) demonstrate that Quizlet integrates notes, spaced repetition, and active learning into an engaging platform, effectively improving vocabulary retention and long-term memory consolidation.

2.2.2.2 | ChatGPT and the Generation of Educational Content

Mishra and Anthony (2023) highlight both the benefits and drawbacks of AI tools like ChatGPT in education, emphasizing the need for responsible integration to enhance critical thinking and active learning processes.

2.2.2.3 | Self-Regulated Learning by Using Study Applications

Baars, Khare, and Ridderstap (2022) demonstrate that study applications with well-designed user interfaces support self-regulated learning by enabling goal-setting, time management, and feedback during study sessions.

2.3 | NLP and Semantic Similarity

The development of NLP techniques has significantly advanced models that analyze and quantify semantic similarity.

2.3.1 | Word Embedding Foundations

The evolution of neural word embeddings represents a foundational shift in computational linguistics, establishing the vector space principles that underpin contemporary sentence embedding methodologies. Church (2017) demonstrates that Word2Vec's widespread adoption stems not from theoretical complexity, but from its practical accessibility and the availability of downloadable implementations that enable immediate application. The method's core innovation lies in representing words as dense vectors in continuous space, where semantic relationships are preserved through geometric relationships rather than symbolic representations.

Mikolov et al. (2013) introduced the skip-gram architecture that learns word representations by predicting contextual words, establishing contrastive learning principles that would later be adapted for sentence-level embeddings. The famous analogy *man : woman :: king : queen* illustrates how vector arithmetic captures semantic relationships, with the mathematical operation $\vec{king} - \vec{man} + \vec{woman} \approx$

quēen demonstrating the foundation of semantic similarity computation through vector mathematics.

Jatnika, Bijaksana, and Suryani (2019) provide empirical validation of Word2Vec's semantic similarity capabilities through extensive evaluation on WordSim-353 and SimLex-999 datasets. Their analysis reveals that cosine similarity, computed as the angular distance between word vectors, effectively quantifies semantic relationships with correlation scores reaching 0.665 on WordSim-353. This establishes the mathematical foundation for similarity measurement that extends to sentence-level embeddings, where the principle that semantically similar entities should be positioned closer in vector space forms the theoretical basis for contrastive learning objectives used in SimCSE and similar frameworks.

2.3.2 | Text Semantic Similarity

Corley et al. (2005) provided foundational insights into computational methods for evaluating semantic similarity, focusing on techniques that measure the closeness of meaning between texts.

2.3.3 | The BERT Framework

Devlin et al. (2019) emphasize that BERT leverages bidirectional representations of language to capture semantic relationships through its transformer-based architecture.

2.3.3.1 | SimCSE

Gao, Yao, and Chen (2021) developed SimCSE, a fine-tuning approach that enhances the quality of sentence embeddings, making them suitable for tasks like semantic similarity evaluation and textual entailment.

SimCSE addresses a fundamental challenge in sentence embeddings known as representation collapse, where different sentences produce nearly identical vector representations, rendering similarity measurements meaningless. Traditional BERT embeddings suffer from this phenomenon due to anisotropic distribution of representations in vector space, where embeddings cluster in narrow regions rather than utilizing the full dimensional space effectively.

The theoretical contribution of SimCSE lies in transforming anisotropic embedding distributions into more isotropic ones. Anisotropic embeddings concentrate in narrow cones within the vector space, limiting their ability to capture fine-grained semantic

distinctions necessary for educational assessment. The contrastive learning framework regularizes the embedding space to achieve uniform distribution, where sentences with different meanings occupy distinct regions while semantically similar sentences remain proximate.

Unlike previous approaches requiring complex data augmentation techniques such as back-translation, paraphrasing, or syntactic transformations, SimCSE achieves superior performance using only dropout noise as minimal data augmentation. This approach preserves semantic content while introducing controlled perturbations necessary for contrastive learning. When the same sentence passes through the encoder twice with different dropout masks, the resulting embeddings form natural positive pairs without external modification.

The mathematical foundation of SimCSE rests on achieving uniform distribution property in the embedding space. This property ensures that the similarity function can effectively discriminate between sentences with varying degrees of semantic overlap, eliminating the bias present in anisotropic embeddings where high similarity scores may result from distributional artifacts rather than genuine semantic relationships.

2.3.3.2 | Contrastive Learning

SimCSE's unsupervised framework generates "positive pairs" by applying independent dropout masks to the same input sentence. Specifically, when a sentence is passed through the BERT encoder twice, the randomness introduced by dropout produces two slightly different embeddings.

SimCSE's contrastive learning framework builds upon foundational principles established in Word2Vec's skip-gram model while extending them to sentence-level representations. The approach creates positive pairs through minimal data augmentation and efficiently generates negative samples through in-batch negatives, eliminating the computational overhead of explicitly creating negative pairs while ensuring diverse negative samples.

For a given sentence s_i , all other sentences s_j where $j \neq i$ in the batch serve as negatives, creating rich contrastive signals without additional data preprocessing. This in-batch negative sampling strategy demonstrates continuity with Word2Vec's contrastive learning principles, where the fundamental approach of learning through contrast remains consistent across word-level and sentence-level embeddings.

$$\ell_i = -\log \frac{\exp(\text{sim}(h_i, h_i^+)/\tau)}{\sum_{j=1}^N \exp(\text{sim}(h_i, h_j)/\tau)}$$

Figure 2.1: Formula for Contrastive Learning

2.3.3.3 | Cosine Similarity

Cosine similarity quantifies the angular distance between embeddings, ensuring that semantically similar sentences are pulled closer together, while dissimilar ones are pushed apart. The temperature parameter τ sharpens the similarity distribution, controlling the influence of positive and negative pairs during training.

The temperature parameter τ in the contrastive learning formula plays a critical role in controlling the sharpness of the similarity distribution. Lower temperature values create more peaked distributions, emphasizing high-confidence similarities, while higher values produce smoother distributions. The optimal temperature balances discriminative power with stability, preventing over-confident predictions that could lead to poor generalization in educational assessment contexts.

$$\text{sim}(h_1, h_2) = \frac{h_1^\top h_2}{\|h_1\| \|h_2\|}$$

Figure 2.2: Formula for Cosine Similarity

2.4 | Progress Tracking

Progress tracking refers to systems that enable users to monitor and evaluate their advancement toward specific goals.

2.4.1 | Progress Tracking as Part of Gameful Design

Deterding et al. (2011) emphasize the importance of structured systems that guide users through progressive goals and stages of development, helping maintain motivation and engagement.

2.4.1.1 | The Three-Star Scoring System

Gastón and Cooper (2017) explore the efficacy of the three-star scoring system, highlighting its ability to improve user engagement and performance in educational and gamified applications.

2.5 | Web Development Framework Literature

The selection of appropriate web development technologies is crucial for building scalable, maintainable, and user-friendly educational applications. This section reviews the literature surrounding modern web development frameworks and libraries that form the technical foundation of contemporary educational platforms.

2.5.1 | Next.js and React Framework

Vercel (2023) emphasizes that Next.js provides a production-ready React framework with built-in optimizations, server-side rendering capabilities, and automatic code splitting that enhances application performance and user experience. The framework's hybrid static and server rendering approach enables developers to create fast, SEO-optimized applications suitable for educational platforms requiring both dynamic content and static resources.

React's component-based architecture, as documented by Facebook (2023), facilitates the development of reusable UI components that can be efficiently maintained and scaled. The library's virtual DOM implementation optimizes rendering performance, which is particularly beneficial for educational applications that frequently update content and user interfaces.

2.5.2 | TypeScript Integration

Microsoft (2023) demonstrates that TypeScript's static type checking reduces runtime errors and improves code maintainability in large-scale applications. The type safety provided by TypeScript is particularly valuable in educational software development, where data integrity and reliable user interactions are paramount.

Bierman et al. (2014) highlight TypeScript's gradual typing system, which allows developers to incrementally add type annotations to existing JavaScript codebases, facilitating smoother migration processes and reducing development risks.

2.5.3 | Data Visualization Technologies

Chart.js, as documented by Chart.js Community (2023), provides a flexible and responsive charting library that enables the creation of interactive data visualizations. In educational applications, effective data visualization supports learning analytics by presenting user progress, performance metrics, and learning patterns in comprehensible formats.

Bostock et al. (2011) established that interactive visualizations enhance user engagement and comprehension, particularly in educational contexts where learners benefit from immediate visual feedback on their progress and performance.

2.5.4 | Styling and User Interface

Wind et al. (2023) document Tailwind CSS as a utility-first CSS framework that accelerates development while maintaining design consistency. The framework's approach to styling through utility classes enables rapid prototyping and maintains design system coherence across educational applications.

Adams (2021) argues that consistent and responsive user interfaces are critical for educational applications, as they reduce cognitive load and allow learners to focus on content rather than navigation challenges.

2.5.5 | Database and Backend Technologies

PostgreSQL's robust relational database capabilities, as documented by PostgreSQL Global Development Group (2023), provide the reliability and ACID compliance necessary for educational applications that manage user data, learning progress, and assessment results. The database's support for complex queries and data integrity constraints ensures accurate tracking of educational metrics.

Cloudinary's cloud-based media management platform, according to Cloudinary (2023), enables efficient handling of educational multimedia content, including automatic optimization and responsive delivery that adapts to various devices and network conditions commonly encountered in educational settings.

2.6 | Summary

The studies reviewed in this chapter highlight key advancements in note-taking strategies, memory processes, NLP, and progress tracking. These insights establish a foundation for the proposed system, integrating effective learning strategies and semantic similarity evaluation to optimize recall and knowledge retention.

The table compares various studies based on their focus on specific aspects or features that are relevant to the proposed approach. It highlights which aspects are addressed and identifies gaps that the proposed approach aims to fill by integrating these features into a unified framework.

Aspects/Features:	Integration of Semantic Similarity Models	Progress Tracking	Focus on Knowledge Consolidation, Memory, and Recall	Flexibility for Varied Note Structures (Formatting)	Feedback and Performance Metrics	App with Interactive UI for Enhanced Usability	Use of NLP for Evaluating Semantic Understanding
Studies:							
Stojanovic et al., 2020	✗	✗	✗	✗	✗	✓	✗
Tran et al., 2020	✗	✓	✓	✗	✓	✓	✗
Baars et al., 2003	✗	✗	✓	✗	✗	✓	✗
Proposed Approach	✓	✓	✓	✓	✓	✓	✓

Figure 2.3: Comparison Table of Related Literature

Theoretical Framework

3.1 | Introduction

3.2 | Learning Theory Foundations

3.3 | Semantic Similarity Theoretical Framework

The theoretical foundation of semantic similarity measurement in natural language processing rests on the principle that semantic relationships can be captured through geometric relationships in vector space. This framework establishes the mathematical and computational foundations underlying the Consol system's approach to evaluating recall performance.

3.3.1 | Vector Space Semantics

The vector space model of semantics, pioneered by Salton, Wong, and Yang (1975) and refined through neural embedding approaches, represents linguistic units as points in high-dimensional space. The fundamental assumption is that semantic similarity correlates with spatial proximity, enabling quantitative measurement of meaning relationships through distance metrics.

Church (2017) emphasizes that Word2Vec's success stems from its ability to capture semantic analogies through vector arithmetic, demonstrating that mathematical operations in embedding space correspond to meaningful semantic transformations. The relationship $\vec{king} - \vec{man} + \vec{woman} \approx \vec{queen}$ illustrates how vector mathematics

can encode complex semantic relationships, establishing the foundation for similarity measurement that extends to sentence-level embeddings.

3.3.2 | Uniform Distribution Property

A critical theoretical requirement for effective semantic similarity measurement is the uniform distribution of embeddings in vector space. Gao, Yao, and Chen (2021) demonstrate that anisotropic embedding distributions, where representations cluster in narrow regions, artificially inflate similarity scores regardless of semantic content.

The uniform distribution property ensures that high similarity scores indicate genuine semantic relationships rather than distributional artifacts, that the similarity function maintains discriminative power across diverse semantic contexts, and that false positives resulting from embedding collapse are minimized. Mathematically, uniform distribution maximizes the entropy of the embedding space, where representations utilize the full dimensional capacity available.

3.3.3 | Representation Collapse Prevention

Representation collapse occurs when distinct inputs produce nearly identical embeddings, eliminating the model's ability to discriminate between different semantic content. This phenomenon is particularly problematic in pre-trained language models where embeddings may concentrate in narrow regions of the vector space, making similarity measurements unreliable for educational assessment.

SimCSE addresses representation collapse through its contrastive learning objective, which actively separates embeddings while maintaining semantic relationships. The theoretical mechanism involves distributional regularization where the contrastive loss encourages embeddings to occupy diverse regions of the vector space, anisotropy reduction where the framework transforms concentrated distributions into more uniform ones, and semantic preservation where genuine similarities are maintained despite increased distribution uniformity.

3.4 | Educational Technology Framework

3.5 | Assessment and Feedback Theory

3.6 | Summary

Methodology

4.1 | Research Methodology

4.2 | System Design Methodology

4.2.1 | Encoder Architecture Design

The Consol system implements a sentence embedding architecture based on SimCSE principles, utilizing BERT as the foundational encoder with specific adaptations for educational assessment contexts. The encoder architecture employs BERT-base-uncased with [CLS] token pooling for sentence representation, differing from mean pooling strategies by utilizing BERT's special classification token trained to capture sentence-level information during pre-training.

The architectural choice of [CLS] pooling over mean pooling is motivated by concentrated representation where the [CLS] token is specifically designed to encode sentence-level information, computational efficiency where single token extraction requires less processing than mean aggregation, and empirical performance where previous studies demonstrate superior results for similarity tasks.

4.2.2 | Temperature Parameter Configuration

The temperature parameter τ in the contrastive learning formula requires careful calibration to achieve optimal similarity discrimination. The parameter controls the sharpness of the similarity distribution, with implications for both precision and recall in educational assessment. The Consol system implements $\tau = 0.05$ based on empirical

validation across diverse educational content domains, balancing discriminative power with generalization ability while preventing over-confident predictions that could lead to poor assessment reliability.

4.3 | Development Process

4.4 | Data Collection Methods

4.5 | Evaluation Framework

4.5.1 | SimCSE Similarity Threshold Calibration

One of the critical aspects of developing Consol was establishing appropriate similarity thresholds for the star rating system that would provide meaningful and fair assessment of student recall performance. This required systematic validation of SimCSE's behavior across various linguistic scenarios commonly encountered in educational contexts.

The initial threshold hypothesis was established based on educational assessment principles:

- 3 stars (Excellent): Similarity ≥ 0.81 - Demonstrates comprehensive understanding
- 2 stars (Good): Similarity ≥ 0.60 - Shows adequate recall with minor gaps
- 1 star (Basic): Similarity ≥ 0.30 - Indicates partial understanding
- 0 stars (Poor): Similarity < 0.30 - Requires significant improvement

To validate these thresholds, we designed a comprehensive test suite encompassing various linguistic transformations that students might employ during recall sessions. The test cases were selected to represent common scenarios in educational recall, including paraphrasing, summarization, expansion, and potential edge cases that could lead to misleading similarity scores.

4.5.2 | Test Case Design and Implementation

Seven representative test cases were developed and evaluated to assess SimCSE performance across different linguistic scenarios. Each test case consisted of an original

sentence (S1) and a student response variant (S2), with expected similarity ranges based on semantic preservation and educational value.

Figure 4.1 presents the complete set of test cases used for threshold validation, while Table 4.1 summarizes the quantitative results and analysis.

The test cases were specifically designed to address:

1. **Semantic Opposition:** Testing SimCSE's ability to detect contradictory meanings
2. **Paraphrasing Detection:** Evaluating performance on synonym-based rewording
3. **Content Expansion:** Assessing similarity when students provide additional detail
4. **Content Summarization:** Testing condensed versions of original content
5. **False Similarity Traps:** Identifying cases where word overlap doesn't indicate understanding
6. **Structural Variations:** Examining different sentence constructions with preserved meaning
7. **Formatting Robustness:** Testing resilience to punctuation and formatting changes

This systematic approach ensured that the final threshold settings would be both educationally meaningful and technically robust, providing students with fair and consistent assessment of their recall performance.

4.5.3 | Validation Results and Threshold Refinement

The systematic testing revealed both strengths and areas for improvement in SimCSE's performance for educational assessment. Analysis of the seven test cases provided crucial insights into the model's behavior across different linguistic scenarios.

Key Findings:

- **False Similarity Detection:** SimCSE successfully identified semantically different sentences with common words (Test 5: 0.27 score), validating its ability to prevent false positive assessments.
- **Structural Robustness:** The model demonstrated strong performance with different sentence structures expressing the same concept (Test 6: 0.88 score), indicating reliable semantic understanding.

1. Simple Opposite Meanings ■ S1: <i>The apple is green.</i> ■ S2: <i>The apple is red.</i> ■ Expected Score: 0.40 - 0.60 2. Paraphrased Sentences (Synonyms & Rewording) ■ S1: <i>She quickly finished her homework.</i> ■ S2: <i>She completed her assignment in no time.</i> ■ Expected Score: 0.75 - 0.95 3. Sentence Expansion (Short to Long Text) ■ S1: <i>The cat sat on the mat.</i> ■ S2: <i>A small, fluffy cat was resting on the comfortable mat by the fireplace.</i> ■ Expected Score: 0.65 - 0.85 4. Summarization (Long to Short Text) ■ S1: <i>The city of Rome, known for its rich history and cultural heritage, is home to iconic landmarks such as the Colosseum and Vatican City, attracting millions of tourists every year.</i> ■ S2: <i>Rome is a historic city with famous landmarks and many visitors.</i> ■ Expected Score: 0.70 - 0.85 5. Common Words, Different Meaning (False Similarity Trap) ■ S1: <i>She runs a software company.</i> ■ S2: <i>He runs five miles every morning.</i> ■ Expected Score: 0.30 - 0.50 6. Different Sentence Structures (Same Idea, Different Style) ■ S1: <i>The storm caused massive destruction in the city.</i> ■ S2: <i>A devastating storm wrecked the city.</i> ■ Expected Score: 0.75 - 0.95 7. Adding Markdown & Symbols (Effect on Similarity) ■ S1: <i>The quick brown fox jumps over the lazy dog.</i> ■ S2: <i>The quick brown fox jumps over the lazy dog!</i> ■ Expected Score: 0.85 - 0.98
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Figure 4.1: SimCSE Test Cases for Threshold Validation: Seven representative scenarios designed to evaluate semantic similarity detection across common linguistic variations in educational contexts.

Table 4.1: SimCSE Threshold Validation Results: Comparison of expected vs. actual similarity scores across seven test scenarios

Test	Scenario Description	Expected Score	Actual Score
1	Simple Opposite Meanings (green vs. red apple)	0.40 - 0.60	0.8988
2	Paraphrased Sentences (homework completion)	0.75 - 0.95	0.6628
3	Sentence Expansion (cat on mat elaboration)	0.65 - 0.85	0.8407
4	Summarization (Rome city description condensed)	0.70 - 0.85	0.8295
5	False Similarity Trap (different meanings of "runs")	0.30 - 0.50	0.2707
6	Different Sentence Structures (storm destruction)	0.75 - 0.95	0.8846
7	Markdown & Symbols (punctuation effects)	0.85 - 0.98	0.9750

Note: Expected scores represent anticipated similarity ranges based on semantic preservation and educational assessment principles. Actual scores reflect SimCSE model output during systematic testing phase.

- **Formatting Resilience:** Punctuation and formatting changes showed minimal impact on similarity scores (Test 7: 0.98 score), ensuring consistent assessment regardless of student writing style.
- **Paraphrasing Sensitivity:** Synonym-based paraphrasing yielded lower scores than anticipated (Test 2: 0.66 vs expected 0.75-0.95), suggesting the need for threshold adjustment to accommodate valid rewording.
- **Opposite Meaning Handling:** Contrary to expectations, semantically opposite statements received high similarity scores (Test 1: 0.90 vs expected 0.40-0.60), indicating a limitation in contradiction detection.

These findings informed the refinement of similarity thresholds and provided validation that SimCSE could serve as a reliable foundation for educational assessment, while highlighting specific scenarios requiring careful interpretation in the final system implementation.

4.6 | Implementation Tools and Technologies

4.7 | Testing and Validation Approach

Results and Discussion

- 5.1 | System Implementation Results**
- 5.2 | Performance Analysis**
- 5.3 | Similarity Scoring Validation Results**
- 5.4 | User Experience Testing Results**
- 5.5 | Learning Analytics Dashboard Results**
- 5.6 | Discussion of Findings**
- 5.7 | Limitations and Challenges**

Conclusions

This section should have a summary of the whole project. The original aims and objective and whether these have been met should be discussed. It should include a section with a critique and a list of limitations of your proposed solutions. Future work should be described, and this should not be marginal or silly (e.g. add machine learning models). It is always good to end on a positive note (i.e. 'Final Remarks').

6.1 | Achieved Objectives

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.2 | Critique and Limitations

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This

text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.3 | Future Work

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

6.4 | Final Remarks

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Resource Persons

Full Name:

Profession:

Office:

Number:

Email Address:

Personal Vitae

- **Full Name:** Mr. George Christopher C. Daguman
- **Residence Address:** Purok-10 Banadero, Ozamiz City, Misamis Occidental
- **Contact Number:** +63 9305976770
- **Email Address:** georgechristopher.daguman@g.msuiit.edu.ph

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